Robust CFAR Detector Based on Truncated Statistics for Polarimetric Synthetic Aperture Radar

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***Abstract*— Constant false alarm rate (CFAR) algorithms using a local training window are widely used for ship detection with synthetic aperture radar (SAR) imagery. However, when the density of the targets is high, such as in busy shipping lines and crowded harbors, the background statistics may be contaminated by the presence of nearby targets in the training window. Recently, a robust CFAR detector based on truncated statistics (TS) was proposed. However, the truncation of data in the format of polarimetric covariance matrices is much more complicated with respect to the truncation of intensity (single polarization) data. In this article, a polarimetric whitening filter TS CFAR (PWF-TS-CFAR) is proposed to estimate the background parameters accurately in the contaminated sea clutter for PolSAR imagery. The CFAR detector uses a polarimetric whitening filter (PWF) to turn the multidimensional problem to a 1-D case. It uses truncation to exclude possible statistically interfering outliers and uses TS to model the remaining background samples. The algorithm does not require prior knowledge of the interfering targets, and it is performed iteratively and adaptively to derive better estimates of the polarimetric covariance matrix (although this is computationally expensive). The PWF-TS-CFAR detector provides accurate background clutter modeling, a stable false alarm property, and improves the detection performance in high-target-density situations. RadarSat2 data are used to verify our derivations, and the results are in line with the theory.**

***Index Terms*— Constant false alarm rate (CFAR), polarimetric whitening filter (PWF), sea clutter, ship detection, statistical modeling, synthetic aperture radar (SAR), truncated statistics (TS).**

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I. Introduction

POLARIMETRIC synthetic aperture radar (PolSAR) is a multidimensional remote sensing system which has been largely used for marine surveillance, including ship detection [1]. The current challenges in ship detection mostly lie in two aspects: the first is to detect small ships in densely packed inshore regions and the second is to detect ships during the medium or high sea states. This article is concerned with the first challenge.

A. Review of Ship Detection in PolSAR Imagery

In SAR images, a common feature of ships is a relatively large backscattering signal, which is usually brighter than the sea's background. This led to designing algorithms that detect vessels using a statistical test on the intensity of the sea clutter [1]. If the statistical distributions of clutter and targets are known, an optimal detector can be designed via the likelihood ratio test (LRT). The optimal polarimetric detector (OPD) was proposed by Novak *et al.* [2]. Polarimetric decompositions and other physically based models [3] have also been adopted for ship detection in PolSAR imagery. Ringrose used the Cameron decomposition method to detect ships from Spaceborne Imaging Radar-C (SIR-C) data [4]. Chen *et al.* [5] proposed a polarization cross-entropy for detecting ships. Zhang *et al.* [6] used the space relation characteristics of ships to detection targets. It has been shown that ship wakes can help detect ships and also provide useful information about heading and velocity [7]–[10]. However, ship wake detection is influenced by various factors, such as radar frequency, angle of view, and angle of incidence and sea state.

The statistical distribution or characteristics of ship backscattering is difficult to obtain, since this depends highly on the physical characteristics of the vessels themselves. Recently, a polarimetric notch filter (PNF) was proposed, which tries to separate ships and vessels based on their polarimetric behavior [11], [12]. It does not assume prior information about the ships and can provide good performance by minimizing the sea clutter power [1], [13], [14]. Another filter working in the absence of the ship's prior information is the polarimetric whitening filter (PWF) proposed by Novak and Burl [15] and Novak *et al.* [16]. The PWF utilizes the polarimetric information to minimize the statistical

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variation due to speckles. In earlier experiments, its performance has proven to be the closest to that of the OPD [15], [16]. The PWF combines all scattering vector correlation information into a single quantity, the scattering matrix. This matrix is not adequate for the representation of the filtered scene received wave, which should be partially polarized because of the spatial scene signal variations within the processing window. Novak's PWF might be used to facilitate image interpretation, but results obtained should be considered with caution because of the eventual loss of polarimetric information [17]. The PWF was further extended to multilook PWF (MPWF) in terms of a polarimetric covariance matrix by Lopes and Sery [18] and Liu *et al.* [19], which makes it consistent with the principle of speckle filtering introduced by Touzi *et al.* [17]. Since the MPWF and the PWF share the same theoretical background in this article, when we refer to PWF we consider both of them depending on an average of whether we perform multilook processing or not.

Both the PNF and PWF are based on the polarimetric covariance matrix, but Touzi *et al.* [20] showed that the covariance matrix and the tools based on it, such as PNF and PWF detectors, cannot cover the full polarimetric information required for enhanced ship detection. Touzi and Vachon [21] also showed that excursion of the degree of polarization provides important information that is not exploited by the covariance matrix as well as the Marino detector. The degree of polarization provides very important information that is not covered by conventional covariance matrix optimization tools such as the Notch detector.

Since we do not have prior information about ships, the test is set on the clutter distribution and, therefore, it considers the probability of false alarms (PFAs). The constant false alarm rate (CFAR) tries to keep the PFA constant by selecting a suitable threshold that adapts with the background clutter [1]. To adapt the threshold, an accurate modeling and estimation of the statistical distribution of local background clutter is generally done using a sliding window [1], [13], [14]. Different statistical models are used to fit heterogeneous sea clutter [22]. In practice, in high-target-density situations such as busy shipping lines and crowded harbors, the training area used to estimate clutter characteristics is generally contaminated by the presence of other targets [23]. This problem was already solved in a single polarization case [23], but there is still no solution when using PolSAR data. The aim of this article is to derive a PolSAR CFAR algorithm which is robust against statistical contamination in the training window. After the target contamination is removed, a statistically robust methodology is presented to model the remaining sea clutter pixels. The PWF is used to produce the image that is truncated. In the following section, a brief review on CFAR detectors in contaminated background is presented.

B. Survey of CFAR Detectors in the Contaminated Surroundings

1) Single Polarization Case: The benchmark of CFAR algorithms is the traditional cell-averaging CFAR (CA-CFAR) which averages each pixel inside the reference window [24]. This is appropriate for a homogeneous clutter but it is affected by contamination.

Data ranking or censoring with different restrictions can be used to remove outliers. The greatest-of CFAR (GO-CFAR) [25] and the smallest-of CFAR (SO-CFAR) [26] detectors are both variations of CA-CFAR, which split the reference window into spatial subsets before averaging. The GO-CFAR trains on the clutter in the greatest subset, while the SO-CFAR uses the smallest. On the other hand, the variability index CFAR (VI-CFAR) [27] chooses a particular group of reference pixels adaptively to estimate background statistics. VI-CFAR is expected to degrade the performance when the distribution of the clutter is complicated and it cannot be modeled by a simple spatial division. Another well-known CFAR algorithm is the ordered statistic CFAR (OS-CFAR) which deals with outliers within the reference window [28]. The OS-CFAR has a significantly higher detection rate and lower false alarm rate [29]. The OS-CFAR disadvantages are generating a small loss in the detection rate in homogeneous clutter compared to the CA-CFAR and a high computational cost. As a generalization of the OS-CFAR detector, the trimmed mean CFAR (TM-CFAR) [30] estimates distribution parameters by the mean of a set of rank-ordered values. In addition, the censored mean-level detector (CMLD) [31] is designed to obtain an acceptable performance in the presence of interfering targets, which is adaptive to several clutter environments. Both ranking and censoring techniques are employed in the CMLD. However, the CMLD may lose its robustness and degrade its CFAR properties when prior knowledge of the interfering targets is absent [32].

The last proposed algorithm is the truncated statistic CFAR (TS-CFAR) detector, which is also the base of this article. If the pixel intensity is higher than a specified threshold, the pixel will be excluded from the training samples and truncated data are used to estimate the clutter parameters. This involves using a different probability density function (PDF) when modeling the truncated data. The difference between truncation and censoring is whether the number of the removed pixels is recorded or not. The number of truncated data points remains unknown, while the number of censored data points is stored. When it comes to model the background in a statistically rigorous manner, the distinction turns to be important [23]. In particular, the truncated data can be represented by the truncated versions of the gamma distribution in the single polarization case, which was revealed to be robust for single polarization images [23].

2) Full Polarization Case: Commonly, the problem of contamination for PolSAR data was addressed using two steps: 1) we select a clean patch close to the detection region and use this to estimate clutter parameters and 2) we perform a CFAR using those parameters. This does not assure us the absence of ships in the training region. However, if ships are few and the region is large, the effect of contamination on the statistical model can be ignored as seen in Gao and Shi's study [14]. This procedure requires supervision. Additionally, if the number of outliers is large or if the sea clutter is too heterogeneous, this procedure may fail.

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For polarimetric data, the problem of removing contamination in the local window is still open, since the truncation can only be applied to a 1-D array and not a 3-D one. A humble idea is to process the different PolSAR channels independently and combine them at the end. This will result in losing polarimetric information since the complex correlation between channels will be ignored.

In this article, we propose a solution based on the PWF. The PWF transforms a matrix into an intensity [15], [16] which is formally equivalent to using a single channel.

C. Contributions of This Study

The statistics of the PWF in Wishart's case have already been derived [22], [33]. Here, we will extend this to the truncated expression. The more challenging issue is to accurately estimate the original polarimetric covariance matrix starting from the truncated data. In other words, we need to find a relationship between the covariance matrix after truncation and the true covariance matrix. The starting point is the relationship between the second moments of a truncated vector and the true covariance matrix derived by Tallis [34]. Unfortunately, this relation assumes that the vector should be real. In this article, we have extended the theorem to the complex case and multilook case. The proof is presented in the Appendix.

In detail, the truncated gamma distribution and the elliptically truncated Wishart's distribution are proposed to model the truncated PWF samples. A statistically rigorous approach is adopted to handle the truncated data in the PWF-TS-CFAR. We use the assumption that the speckle is fully developed and that the radar cross section (RCS) is locally constant, which makes the problem mathematically tractable. It is well known that this assumption may not be sustained as the SAR resolution increases and the sea state roughens. Fortunately, this assumption should cover most of the high-target-density situations, which mostly happen in sheltered areas (e.g., harbors). In addition, a mixture of several gamma distributions can also be used to model large-scale heterogeneity [35]. Finally, since the PWF distribution was found to be a gamma and this is a special case of the generalized gamma distribution (GGD), we decided to improve the robustness of the statistical model by adding an expression which considers the GGD.

This article is organized as follows. Section II presents the statistical PolSAR data involving the MPWF. Section III provides an introduction to TS for a matrix and derives the proposed TS-CFAR detectors for PolSAR measurements. In addition, the performance of parameter estimation and the false alarm regulation property and the receiver operating characteristic (ROC) of the CFAR detector are examined based on simulated measurements of sea clutter drawn from the truncated Wishart distribution. In Section IV, an adaptive truncated threshold is presented, and the PWF-TS-CFAR detector is compared among different parameter estimators via real Radarsat-2 SAR measurements. Finally, Section V presents the main conclusion and perspectives.

II. Statistical Model of PolSAR Data

A. Description of PolSAR Data

Under far-field hypothesis, the scattering characteristics of a target are represented by a polarimetric scattering matrix, expressed in the following equation when a horizontal–vertical linear basis is used [1], [15]:

$$\mathbf{S} = \begin{bmatrix} S_{\mathrm{HH}} & S_{\mathrm{HV}} \\ S_{\mathrm{VH}} & S_{\mathrm{VV}} \end{bmatrix} \tag{1}$$

where S_{xy} represents the complex scattering coefficient with x standing for the transmitting polarization, and y for the receiving polarization (H, horizontal polarization and V, vertical polarization). When the system is monostatic and the reciprocity limitation is satisfied, we have $S_{\mathrm{HV}} = S_{\mathrm{VH}}$. Then, the scattering vector can be defined as

$$\mathbf{k} = \begin{bmatrix} S_{\mathrm{HH}} \\ \sqrt{2}S_{\mathrm{HV}} \\ S_{\mathrm{VV}} \end{bmatrix}. \tag{2}$$

To reduce speckles in an SAR image, filtering or multilook processing can be used. We can therefore obtain the covariance matrix as [15], [19]

$$\mathbf{C} = \frac{1}{L}\sum_{i=1}^{L} \mathbf{k}_i \mathbf{k}_i^{\dagger} \tag{3}$$

where L is the number of averaged looks. The superscript $\dagger$ denotes the conjugate transpose. The multilook covariance matrix $\mathbf{C}$ is a random variable and therefore it can be modeled using a PDF. $\mathbf{\Sigma} = E\{\mathbf{C}\}$ is the covariance matrix of the speckle in the Gaussian case, $\mathbf{\Sigma}$ is the statistical mean of the multilook covariance matrix $\mathbf{C}$, and $E(\cdot)$ represents the expectation operator.

One part of the SAR pixel variation is due to the interference between scatters within a resolution cell, which is also referred to as speckle. The other part of the variation is due to the fluctuation of the RCS, which is also referred to as texture [14]. The variation generated by texture is slower than the one generated with speckles. In other words, although the texture is modulated, it can only be observed on a large amount of pixels. The PDF of a covariance matrix without texture is a Wishart distribution based on multivariate complex Gaussian distribution, which models pure and fully developed speckles [36], [37]

$$f_C(\mathbf{C}) = \frac{L^{Ld}|\mathbf{C}|^{L-d}\exp(-L\mathrm{Tr}(\mathbf{\Sigma}^{-1}\mathbf{C}))}{\Gamma_d(L)|\mathbf{\Sigma}|^L} \tag{4}$$

where L is the number of looks, d is the dimension of $\mathbf{C}$, $\mathrm{Tr}(\cdot)$ is the trace operator, $\Gamma(\cdot)$ is the gamma function, and $\Gamma_d(L)$ is

$$\Gamma_d(L) = \pi^{\frac{1}{2}d(d-1)}\Gamma(L)\cdots\Gamma(L-d+1). \tag{5}$$

B. Statistics of the MPWF

The different polarimetric channels can be processed together to reduce speckles [19]. Novak and Burl [15] proposed the PWF which is an optimization aimed at maximally

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reducing speckles. Novak *et al.* [16] later showed that the PWF can improve the target detection performance. The PWF was further enhanced into an MPWF by Lopes and Sery [18] and Liu *et al.* [19]. PWF can be seen as a generalization of the MPWF.

The expression of the MPWF for L looks is [15]

$$z = \frac{1}{L}\sum_{i=1}^{L} \mathbf{k}_i^{\dagger}\boldsymbol{\Sigma}^{-1}\mathbf{k}_i = \mathrm{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{C}) \tag{6}$$

where $\mathrm{tr}(\cdot)$ is the trace operator and L is the number of looks.

When speckle is fully developed as mentioned in Section II-A, z obeys a gamma distribution after a series of mathematical deduction [18], [33]

$$z \sim \gamma\left(Ld, \frac{1}{L}\right) \tag{7}$$

where $\gamma(\alpha, \beta)$ denotes a gamma distribution with shape parameter α and scale parameter β. $\gamma(\alpha, \beta)$ can be represented as follows [38]:

$$\gamma(\alpha, \beta) = \frac{1}{\Gamma(\alpha)}\frac{1}{\beta}\left(\frac{x}{\beta}\right)^{\alpha-1}\exp\left(-\frac{x}{\beta}\right) \tag{8}$$

where $\Gamma(a) = \int_0^{\infty} y^{a-1}e^{-y}dy$ is the gamma function.

III. Polarimetric Truncated Statistics CFAR Detector

A. Truncated Statistics

Clutter parameter estimation in areas contaminated by outliers (or interfering targets) is challenging. In this article, we will use truncation since it allows deriving a rigorous statistical model.

The PDF of the truncated data can be calculated starting from the PDF of the untruncated data.

Let us assume the random variable z can be modeled by the PDF $f_z(z)$ and a cumulative distribution function (CDF) $F_z(z)$. Suppose $\tilde{z}$ be the truncated version of z after the truncation by a threshold ρ. The right truncated distribution can be defined as

$$f_{\tilde{z}}(z;\rho) = f_z(z|z<\rho) = \begin{cases} \dfrac{f_z(z)}{F_z(\rho)}, & 0 < z \le \rho \\ 0, & x > \rho. \end{cases} \tag{9}$$

The division by $F_z(\rho)$ ensures that the integral of $f_{\tilde{z}}(z;\rho)$ is unitary. The truncation threshold is also referred to as truncation depth. Since there is no prior knowledge about outliers, it is difficult to estimate the optimal truncation depth. A lower threshold ρ returns a higher probability to exclude all outliers; however, if the truncation threshold is too excessive, inaccurate parameter estimation may occur.

There are several studies that deal with the 1-D case [39], [40]. The first results on truncated moments considered a linear truncated multivariate normal (MVN) distribution [41]. Tallis [34] later extended his results of linear truncations to the case of elliptical and radial truncation. Tallis [42] built on previous results to compute the moments of a normal distribution with a plane truncation. For a review of truncated moments for different continuous distributions, refer to [29] and [44]. The constraint on the elliptical and radial truncation is as follows [34]:

$$0 \le z = x'\boldsymbol{\Sigma}^{-1}x \le \rho \tag{10}$$

where x is a multivariate vector, $'$ is the transpose operator, and z, $\boldsymbol{\Sigma}$, ρ are all real.

Interestingly, this constraint is also the form of the PWF [(6)] when the number of look is $L = 1$. This further motivated our work.

B. PWF-TS-CFAR Detector

In this section, CFAR detectors based on TS are derived for single-look complex (SLC) and multilook complex (MLC) SAR measurements, which are modeled by the gamma and Wishart's distributions. We consider the MLC case first, since the SLC can be seen as a special case of MLC. The intensity of PWF pixels is positive and assumed to be independent and identically distributed.

1) Estimation of the Polarimetric Covariance Matrix: The complete and truncated distributions [(7) and (9)] model the PWF properly only in the assumption that the polarimetric covariance matrix is estimated accurately. The problem of estimating the polarimetric covariance matrix from the truncated data is of paramount importance.

As mentioned previously, the MPWF is an elliptical and radial truncation. For fully developed speckle (i.e., Wishart's distribution), the true covariance matrix can be derived using the second truncated moments. From Tallis' study, it turns out that [34]

$$\mathfrak{S}(\rho;\boldsymbol{\Sigma}) = \frac{\boldsymbol{\Sigma}}{\mu_T(\rho)}, \quad \mu_T(\rho) \equiv \frac{F_d(\rho)}{F_{d+2}(\rho)} \tag{11}$$

with $F_v(\cdot)$ denoting the cumulative distribution of a χ^2 variable with d degrees of freedom. Inverting (18) to express $\boldsymbol{\Sigma}$ as a function of $\mathfrak{S}$ is trivial, since $c_T(\rho)$ is a scalar damping factor independent of $\boldsymbol{\Sigma}$. This result is only valid for Gaussian's distributed vectors or covariance matrices.

According to the methods in [34] and [47], and the relationship between the χ^2 cumulative distribution $F_v(\cdot)$ and the incomplete gamma function $\Gamma(a, b)$ [38], we can extended the real number results to the complex and multilook case (see the Appendix)

$$\boldsymbol{\Sigma} = \mu_T(\rho)\mathfrak{S}(\rho;\boldsymbol{\Sigma}), \mu_T(\rho) \equiv \frac{\Gamma(Ld, L\rho)}{\Gamma(Ld+1, L\rho)} \tag{12}$$

with $\Gamma(a, b) = 1/(\Gamma(a))\int_0^b y^{a-1}e^{-y}dy$. Thus, we can estimate the true polarimetric covariance matrix of the clutter from the second moments of the truncated data. This makes the PDF in (12) more accurate.

2) Estimation of the Equivalent Number of Looks: From (12), we can see that the difference between the covariance matrix of the truncated data and the true covariance matrix is just a coefficient. The polarimetric behavior of the clutter is unchanged between the complete and truncated set after the elliptical and radial truncation (PWF). It is clear that if we do not adjust the covariance matrix with the correct coefficient $\mu_T(\rho)$, the output of the MPWF will be different in scale.

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This difference only affects the mean of the MPWF distribution. Therefore, there are two parameters that need to be estimated, the equivalent number of looks (ENL) and the mean of the MPWF. The distribution of the MPWF without covariance truncation is

$$z \sim \gamma\left(Ld, \frac{1}{L}\right). \tag{13}$$

Its CDF is

$$F_z(z; L, d) = \Gamma(Ld, Lz). \tag{14}$$

The PDF of $\tilde{z}$, the truncated data is

$$f_{\tilde{z}}(z; L, d) = \begin{cases} (L)^{Ld}\dfrac{z^{Ld-1}\exp(-Lz)}{\Gamma(Ld)\Gamma(Ld, L\rho)}, & 0 < z \le \rho \\ 0, & z > \rho. \end{cases} \tag{15}$$

The other estimator is based on the moments. The nth order moments of the truncated MPWF in (15) is

$$E_{\tilde{z}}(z^n) = \frac{\Gamma(Ld+n)\Gamma(Ld+n, L\rho)}{L^n\Gamma(Ld)\Gamma(Ld, L\rho)}. \tag{16}$$

Then, we can use the first moments of the truncated data to estimate the ENL

$$\Gamma(x+1) = x\Gamma(x). \tag{17}$$

We can get

$$\langle z\rangle = \frac{d\Gamma(\hat{L}d+1, \hat{L}\rho)}{\Gamma(\hat{L}d, \hat{L}\rho)}. \tag{18}$$

A maximum-likelihood (ML) estimator (MLE) for the ENL can be also obtained from the likelihood function

$$\begin{aligned}\mathcal{L}(L|\tilde{z}) &= \prod_{i=1}^{n} f_{\tilde{z}}(\tilde{z}_i|L) \\ &= (L)^{nLd}\frac{\exp(-L\sum_{i=1}^{n}\tilde{z}_i)}{[\Gamma(Ld)\Gamma(Ld, L\rho)]^n}\prod_{i=1}^{n}\tilde{z}_i^{Ld-1}\end{aligned} \tag{19}$$

where $\{\tilde{z}_i\}_{i=1}^{n}$ is a size n sample of truncated MPWF outputs. The log-likelihood function is derived as

$$\begin{aligned}&\log\mathcal{L}(L|\tilde{z}) \\ &= nLd\log L - n\log\Gamma(Ld, L\rho) \\ &\quad - n\log\Gamma(Ld) - nL\frac{1}{n}\sum_{i=1}^{n}\tilde{z}_i + n(Ld-1)\frac{1}{n}\sum_{i=1}^{n}\log(\tilde{z}_i)\end{aligned} \tag{20}$$

where $(1/n)\sum_{i=1}^{n}\tilde{z}_i$ and $(1/n)\sum_{i=1}^{n}\log(\tilde{z}_i)$ are the statistical means (SMs) of the original and logarithmic truncated MLC measurements. Thus, we can get the MLE estimate of the ENL

$$\widehat{L} \subseteq \arg\max_{L}\{\log\mathfrak{L}(L|\tilde{z})\} \tag{21}$$

which must be solved numerically. Thus, both the ENL and the polarimetric covariance matrix are well estimated.

To summarize, there are two parameters that need to be estimated, the ENL and the mean of MPWF.

The distribution of the MPWF without the calibration of polarimetric covariance matrix is

$$z \sim \gamma\left(Ld, \frac{\mu_T(\rho)}{L}\right). \tag{22}$$

Here, we can denote $\mu = \mu_T(\rho)$ to simplify the notation. The PDF of $\tilde{z}$, the truncated data is

$$f_{\tilde{z}}(z; L, d) = \begin{cases} \left(\dfrac{L}{\mu}\right)^{Ld}\dfrac{z^{Ld-1}\exp\left(-\dfrac{Lz}{\mu}\right)}{\Gamma(Ld)\Gamma\left(Ld, \dfrac{L\rho}{\mu}\right)}, & 0 < z \le \rho \\ 0, & z > \rho. \end{cases} \tag{23}$$

An MLE can be obtained from the likelihood function

$$\begin{aligned}&\log\mathcal{L}(\mu, L|\tilde{z}) \\ &= nLd\log\frac{L}{\mu} - n\log\Gamma\left(Ld, \frac{L\rho}{\mu}\right) - n\log\Gamma(Ld) \\ &\quad - \frac{nL}{\mu}\frac{1}{n}\sum_{i=1}^{n}\tilde{z}_i + n(Ld-1)\frac{1}{n}\sum_{i=1}^{n}\log(\tilde{z}_i).\end{aligned} \tag{24}$$

We can solve the maximization numerically to obtain the best estimations of μ, L.

There are several methodologies that can estimate the ENL in polarimetric data [49]. Once the ENL is estimated it can be considered as an image constant.

We are therefore left with the estimation of the mean value μ. If the ENL is known to us, $\partial/\partial\mu\log\mathcal{L}(\mu, L|\tilde{z}) = 0$ can be used to derive the derivative of the log-likelihood function as

$$f(t) = \frac{t^{Ld}\exp(-t)}{\Gamma(Ld)\Gamma(Ld, t)} - L\left(d - \frac{\tilde{z}t}{L\rho}\right) = 0 \tag{25}$$

where $t = L\rho/\mu$.

We can obtain accurate estimations of ENL by the iterative censoring (IC) scheme. In this case, the truncated radial is fixed, and the pixels of the truncated data are updated with the truncated polarimetric covariance matrix until the difference between the numbers of the truncated pixels in the IC scheme is small enough.

Although the iterative detection procedure may need several cycles and require long calculation time, the IC scheme has shown robust performance in the dense target situation and can be integrated with the CFAR algorithm.

The specified false alarm rate P_{FA} can be related to the CDF, parameterized with the estimated mean value, as

$$P_{FA} = 1 - F_z(T; L, d) = 1 - \Gamma\left(Ld, \frac{LT}{\mu}\right) \tag{26}$$

where T is the detection threshold that needs to be solved

$$T = \frac{\mu}{L}\Gamma^{-1}(Ld, 1-P_{FA}). \tag{27}$$

3) Generalized Model of the TS: Since (13) and (21) represent the gamma distribution which is a special case of the GGD, it is possible to make this expression more robust if we use the GGD instead as

$$\gamma(\alpha, \beta) = f(x; \alpha, 1, \alpha\beta) \tag{28}$$

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where f is the GGD as follows:

$$f(z;k,\upsilon,\sigma)=\frac{|\upsilon|k^k}{\sigma\Gamma(k)}\left(\frac{z}{\sigma}\right)^{k\upsilon-1}\exp\left\{-k\left(\frac{z}{\sigma}\right)^{\upsilon}\right\} \tag{29}$$

where k,υ,σ are the shape, power, and scale parameters, respectively.

$$k=Ld,\sigma=d\mu.$$

Its cumulative density function (CDF) is

$$F_z(z;L,d)=\begin{cases}\Gamma[k,\eta z^{\upsilon}], & \upsilon>0\\ 1-\Gamma[k,\eta z^{\upsilon}], & \upsilon<0\end{cases} \tag{30}$$

where $\eta=k\sigma^{-\upsilon}$. The PDF of $\tilde{z}$, the truncated data is

$$f_{\tilde{z}}(\tilde{z};k,\upsilon,\sigma)=\begin{cases}\dfrac{f(z;k,\upsilon,\sigma)}{F_z(\rho;L,d)}, & 0<z\le\rho\\ 0, & z>\rho.\end{cases} \tag{31}$$

Via the definition of the incomplete gamma function, we can get the nth moments of the truncated data after MPWF

$$E_{\tilde{z}}(\tilde{z}^n)=\frac{\eta^{-\frac{n}{\upsilon}}\Gamma\left(k+\frac{n}{\upsilon}\right)\Gamma\left(k+\frac{n}{\upsilon},\eta\rho^{\upsilon}\right)}{\Gamma(k)\Gamma[k,\eta\rho^{\upsilon}]}. \tag{32}$$

The ML estimator is used

$$\begin{aligned}&\log\mathcal{L}(k,\upsilon,\sigma\,|\tilde{z})\\ &=n\log|\upsilon|+nk\log k+(k\upsilon-1)\sum_{i=1}^{n}\log(\tilde{z}_i)\\ &\quad-nk\upsilon\log\sigma-\frac{k}{\sigma^{\upsilon}}\sum_{i=1}^{n}\tilde{z}_i^{\upsilon}-n\log\Gamma(k)-n\log\Gamma(k,\eta\rho^{\upsilon}).\end{aligned} \tag{33}$$

By differentiating the log-likelihood function with respect to k,υ,σ and setting them to 0, we obtain the following three equations:

$$\begin{aligned}\frac{\partial\log\mathcal{L}}{\partial k}&=n+2n\log\eta+n\upsilon\log\rho-\frac{nK}{\Gamma(k)\Gamma(k,\eta\rho^{\upsilon})}\\ &\quad-\frac{n}{\sigma^{\upsilon}}\left(\frac{1}{n}\sum_{i=1}^{n}\tilde{z}_i^{\upsilon}\right)+n\upsilon\left(\frac{1}{n}\sum_{i=1}^{n}\log\tilde{z}_i\right)=0\end{aligned} \tag{34}$$

$$\begin{aligned}\frac{\partial\log\mathcal{L}}{\partial\upsilon}&=n\frac{1}{\upsilon}+k\sum_{i=1}^{n}\log(\tilde{z}_i)-nk\log\sigma\\ &\quad-n\frac{(\eta\rho^{\upsilon})^k\exp(-\eta\rho^{\upsilon})\log\rho}{\Gamma(k)\Gamma(k,\eta\rho^{\upsilon})}\\ &\quad-n\frac{k}{\sigma^{\upsilon}}\left(\frac{1}{n}\sum_{i=1}^{n}\tilde{z}_i^{\upsilon}\log(\tilde{z}_i)\right)=0\end{aligned} \tag{35}$$

$$\frac{\partial\log\mathcal{L}}{\partial\sigma}=-n\frac{k\upsilon}{\sigma}+\frac{k\upsilon}{\sigma^{\upsilon+1}}\sum_{i=1}^{n}\tilde{z}_i^{\upsilon}+n\frac{(\eta\rho^{\upsilon})^k\exp(-\eta\rho^{\upsilon})\upsilon}{\sigma\Gamma(k)\Gamma(k,\eta\rho^{\upsilon})}=0 \tag{36}$$

where $K=\Gamma(k)\psi(k)-\eta\rho^{\upsilon}T(3,k,\eta\rho^{\upsilon})$, $\eta=k\sigma^{-\upsilon}$, and in [48]

$$T(m,s,z)=G_{m-1,m}^{m,0}\left(\begin{matrix}0,0,\ldots,0\\ s-1,-1,\ldots,-1\end{matrix}\,|x\right). \tag{37}$$

MeijerG function is defined as

$$\begin{aligned}&G_{p,q}^{m,n}\left[z\,\middle|\,\begin{matrix}a_1,\ldots,a_p\\ b_1,\ldots,b_q\end{matrix}\right]\\ &=\frac{1}{2\pi i}\\ &\quad\times\int_L\frac{\prod_{j=1}^{m}\Gamma(b_j-s)\prod_{j=1}^{n}\Gamma(1-a_j+s)}{\prod_{j=m+1}^{q}\Gamma(1-b_j+s)\prod_{j=n+1}^{p}\Gamma(a_j-s)}z^s ds.\end{aligned} \tag{38}$$

The PFA and probability of detection (PD) can be rewritten as

$$P_{FA}=1-F_z(z;L,d)=\begin{cases}1-\Gamma[k,\eta T^{\upsilon}], & \upsilon>0\\ \Gamma[k,\eta T^{\upsilon}], & \upsilon<0\end{cases} \tag{39}$$

The threshold turns out to be

$$T=\begin{cases}\left(\dfrac{1}{\eta}\Gamma^{-1}[k,1-P_{FA}]\right)^{1/\upsilon}\\ \left(\dfrac{1}{\eta}\Gamma^{-1}[k,P_{FA}]\right)^{1/\upsilon}\end{cases} \tag{40}$$

where $\eta=k\sigma^{-\upsilon}$. It can also be seen that the calculation of the threshold is also easier to obtain compared with the gamma distribution.

In the following, we will use both the truncated gamma distribution and the truncated GGD to fit simulated and real PolSAR data.

C. Performance of Parameter Estimation

This section presents the empirical results that: 1) show the correctness of our derived distributions after truncation; 2) evaluate the performance of ENL estimator for truncated data; and 3) compare with the conventional ML estimator for uncontaminated data.

First, we evaluate the correctness of the distribution of the truncated data. Monte Carlo simulations are used to produce simulated SAR data. We used the simulation to control exactly the statistics of clutter and contaminating targets. We generated the Wishart distribution covariance matrices and the textual variables. The Wishart part of the synthetic data set consists of $N=1\,000000$ covariance matrix samples drawn from a complex, circular, and zero-mean Wishart distribution. The distribution was colored by a matrix that is computed by averaging a homogeneous region of sea in real SAR data. The number of looks in the simulation was set to $L=10$. From $N=1\,000\,000$ samples, we drew $M=[1000\ 10\,000\ 100\,000]$) bootstrap samples to evaluate the correctness of the truncated distribution, which are shown in Fig. 1 for (15). It can be seen that the theoretical PDF is quite in accordance with the simulated data, especially when the sample size is large enough. We can get the same results of the other expressions of the truncated PDFs in (23) and (31).

Second, we assess the performance of different ENL estimators by the mean square error (MSE), which is defined as

$$\begin{aligned}\text{Bias}(\hat{\beta})&=|E(\hat{\beta})-\beta|/\beta\\ \text{Var}(\hat{\beta})&=E\{[E(\hat{\beta})-\hat{\beta}]^2\}/\beta^2\\ \text{MSE}(\hat{\beta})&=E[(\hat{\beta}-\beta)^2]/\beta^2=\text{Bias}(\hat{\beta})^2+\text{Var}(\hat{\beta})\end{aligned} \tag{41}$$

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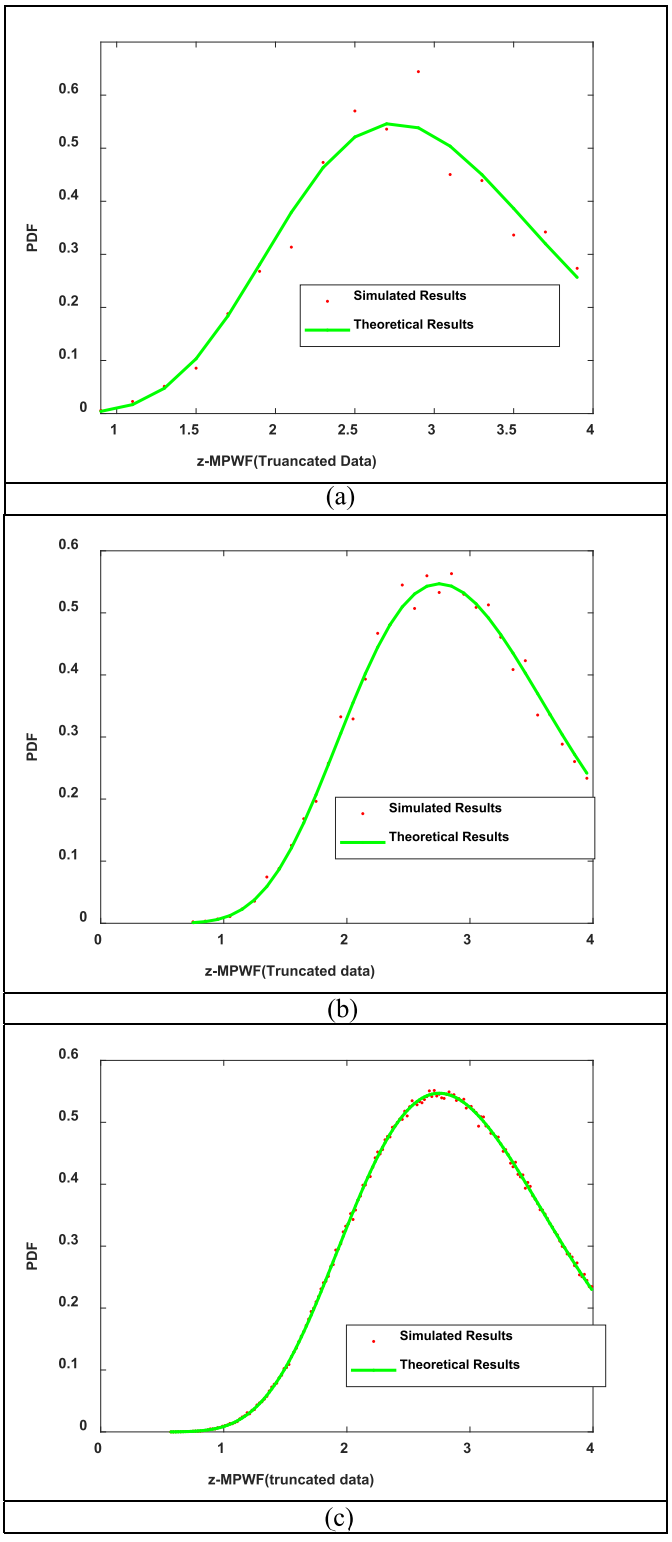


Fig. 1. Validation of the TS. (a) Sample size 1000. (b) Sample size 10 000. (c) Sample size 100 000.

where β is the true value and $\hat{\beta}$ is the estimation of β. To describe 1) the simulation of contaminated data and 2) the depth of PWF truncation, we define two parameters. The contamination ratio R_c is defined as the proportion of contaminated data points relative to the total samples size, and the truncation ratio R_t as the fraction of truncated samples over the total samples size. Here, we set the contamination ratios as $R_c = 0\%$ and $R_c = 20\%$, respectively. And the target-to-clutter ratio (TCR) is defined as TCR $= \mathrm{tr}(\mathbf{\Sigma}_T - \mathbf{\Sigma}_C)/\mathrm{tr}(\mathbf{\Sigma}_C)$, where $\mathbf{\Sigma}_T$ is the polarimetric covariance matrix of target and $\mathbf{\Sigma}_C$ is the polarimetric covariance matrix of sea clutter. The contamination samples are drawn from a gamma distribution TCR $= 2$. Unfortunately, both the estimators based on PWF-TS should solve the complicated transcendental equations numerically. We found a group of ENL estimators based on the submatrix and log-cumulants [49], which may be used in both the untruncated and truncated data perfectly and cost little time.

The results in Fig. 2 are in the case of the uncontaminated data, and the results for the contaminated data are listed in Fig. 2.

In both Figs. 2 and 3, $d1d2$, $d2d3$, $d1d3$, $d1d2d3{-}1$, and $d1d2d3{-}2$ are all estimators based on the submatrices in [49]. From the results, we can see that the performances of the submatrix estimators give the best results and each of them gives similar performance. Therefore, we can take one as the ENL-SM method. The ML method is proposed by Anfinsen, which is used for the untruncated data and can be named as ENL-SN. The ML-truncated method is our method suitable for truncated data, which is named as ENL-ML. It can be seen when the truncated ratio almost equals the contaminated ratio, the ML and the ML-truncated methods may give a good performance.

If the ENL has been obtained, then $\mu_T(\rho)$ can be obtained easily via (12) or the ML estimator in (25). We also found that the solutions via (24) and (33) are almost the same as the above, which has also been verified by the measured data in Section IV. Therefore, we can choose suitable methods for parameter estimations considering both the accuracy and the efficiency.

D. Assessment of the PWF-TS-CFAR Detector

In this section, we use two CFAR indexes to assess the performance of the PWF-TS-CFAR detector: 1) the false alarm rate maintenance and 2) the ROC. We calculate these indexes from the simulated gamma-distributed clutter. The proposed PWF-TS-CFAR detector is compared with the conventional TS-CFAR detectors based on intensity only.

1) False Alarm Rate Maintenance: The observed false alarm rate is defined as

$$P_{fa} = \frac{n_{fa}}{n} \tag{42}$$

where n_{fa} is the number of false alarms and n is the total number of samples. The specified false alarm rate is denoted as P_{FA}. A constant P_{fa} can be approached to P_{FA} in theory if the statistical model of clutter satisfies the assumed model and the corresponding parameter estimates are accurate. Therefore, we define the false alarm rate maintenance

$$C_L = 10\log\left(\frac{P_{fa}}{P_{FA}}\right). \tag{43}$$

This is a fundamental parameter to assess the CFAR maintenance.

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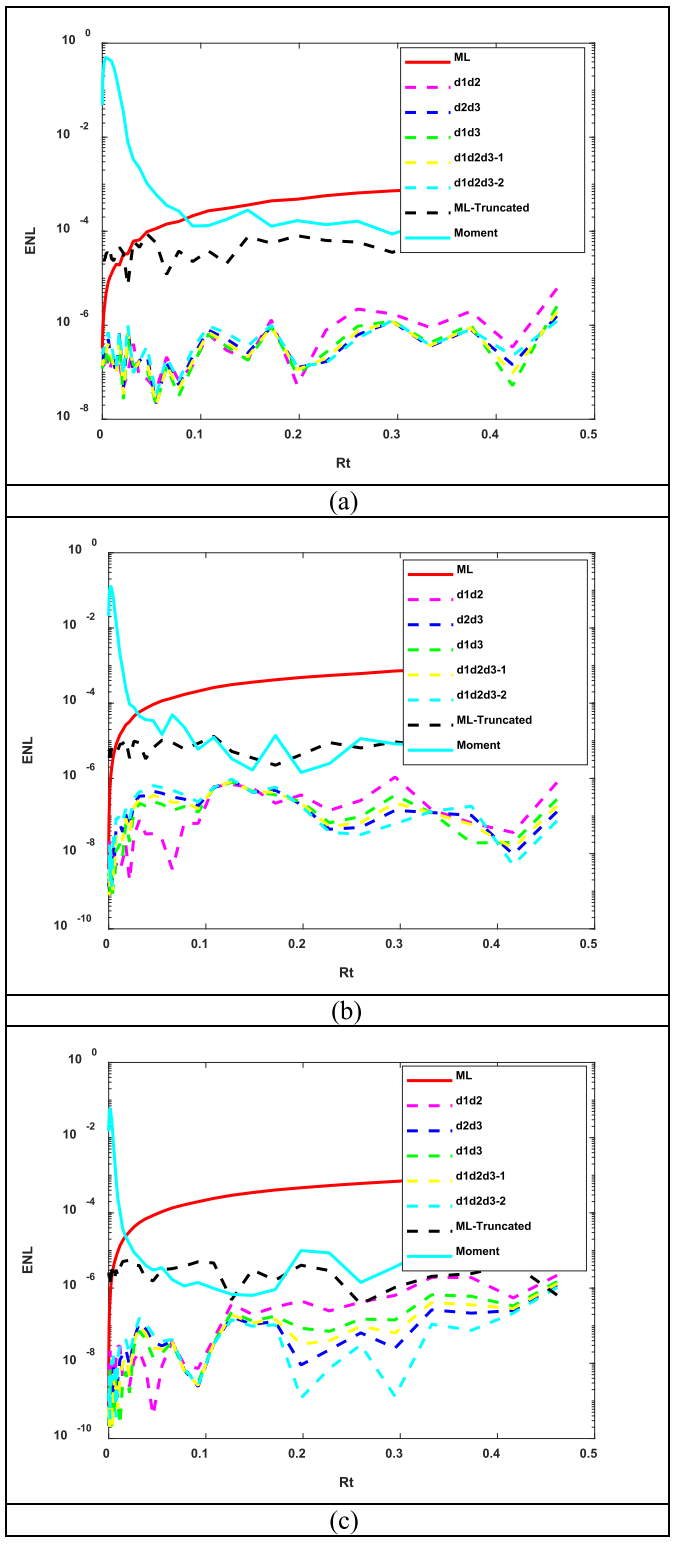


Fig. 2. Performances of different ENL estimators with $R_c = 0$ and TCR $= 2$. (a) Sample size 1000. (b) Sample size 5000. (c) Sample size 10 000.

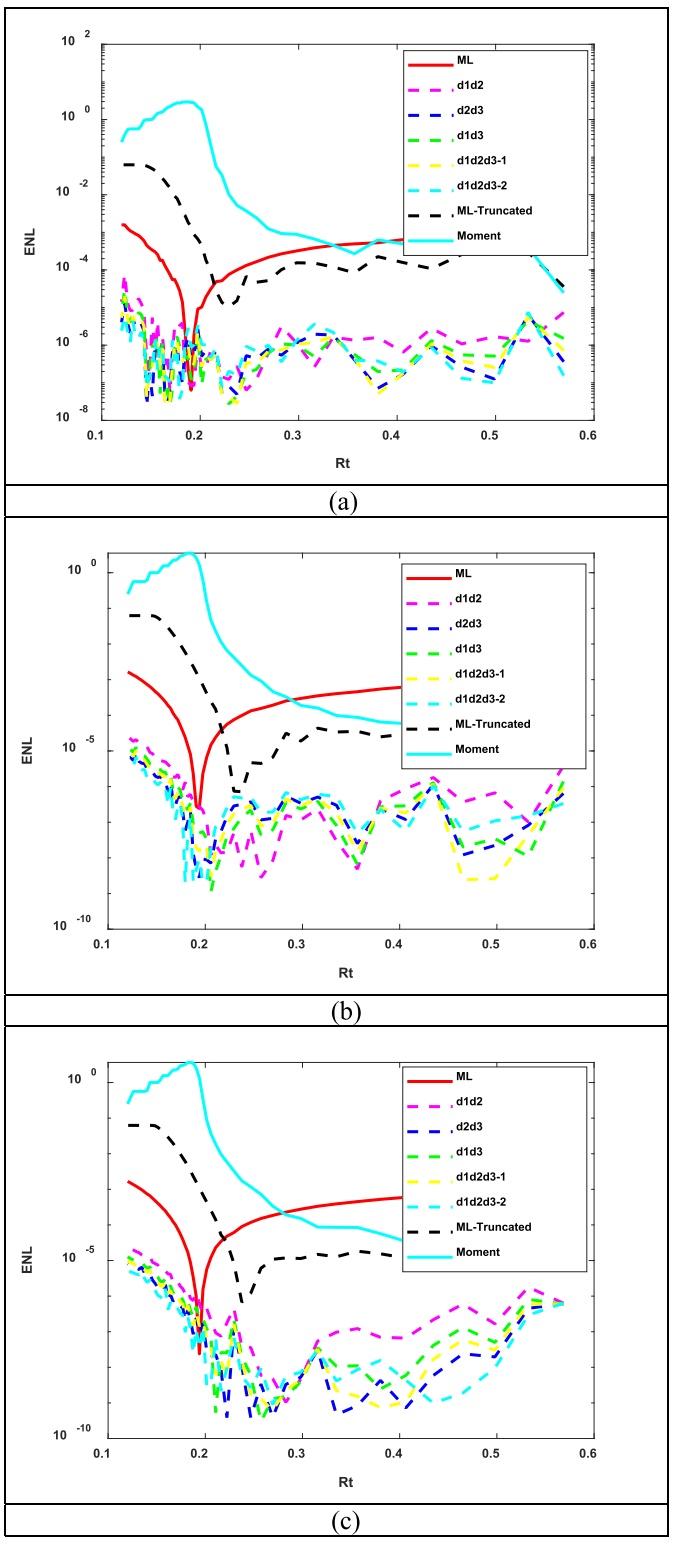


Fig. 3. Performances of different ENL estimators with $R_c = 20\%$ and TCR $= 2$. (a) Sample size 1000. (b) Sample size 5000. (c) Sample size 10 000.

2) ROC: The detection rate is measured as

$$P_d = \frac{n_d}{n_t} \tag{44}$$

where n_d is the number of the detected targets and n_t is the total number of targets. P_d is usually evaluated at different values of P_{FA} to assess detection performance, where P_{FA} is chosen that produces a P_{fa} that meets the practical requirements. This forms the ROC curve and it is always used to assess different detector performance. We use Monte Carlo simulations to draw the ROC curve. Here, the sample size is set to 1000, which simulates the samples in a reference window.

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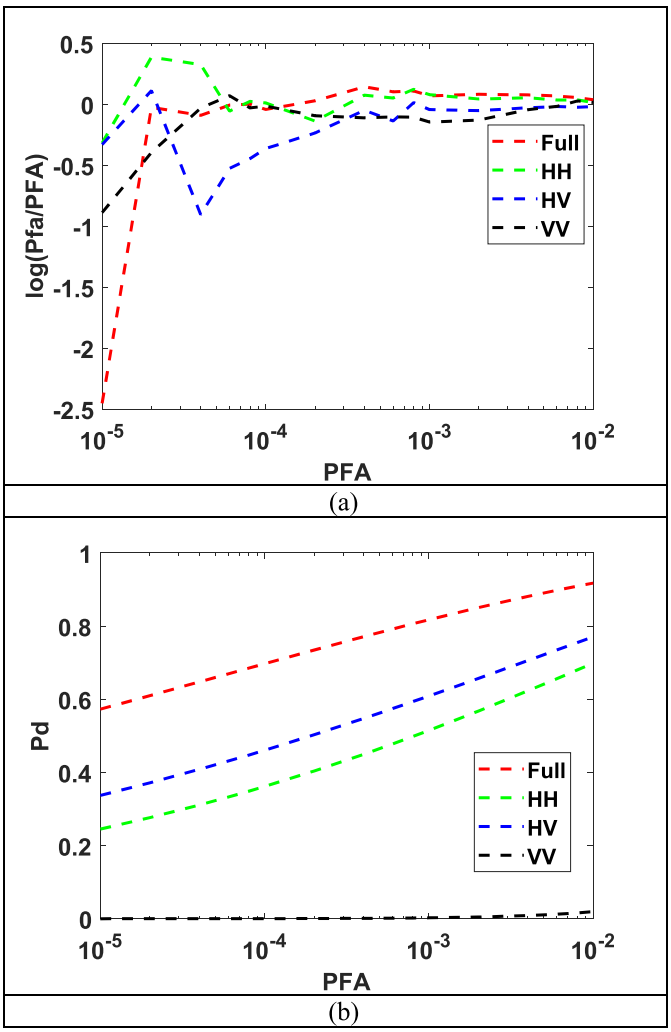


Fig. 4. Assessment of the Pol-TS-CFAR. (a) Bias of the PFA from the true value PFA. (b) ROC curves when TCR = 2.

The contamination rate is set as $R_c = 20\%$, and the truncation ratio $R_t = 20\%$. The total number of simulations is 10 000, which can make P_{fa} and P_d almost the same as the true values.

The results are shown in Fig. 4. P_{fa} almost equals the PFA, which means that the truncated parameter estimation is correct and efficient. The biases of P_{fa} for all truncated data in Fig. 4(a) is near to each other and give the similar maintenance of the CFAR for values of PFA larger than 2e − 5. Fig. 4(b) shows that the full polarimetric detection gives the best performance since all polarimetric information is used. It can also be seen that the HV polarization channel is the best one for detection among all the three single polarizations. The phenomenon that the VV polarization gives the worst result shows that the VV intensity from the targets is similar to that from the clutter, leading to the worst detection performance.

IV. Detection Performance With Real Data

The workflow of the PWF-TS-CFAR detector proposed in this article is shown in Fig. 5. There are three important processors, which include the local detection design, the data truncation processor, and the CFAR processor. These are

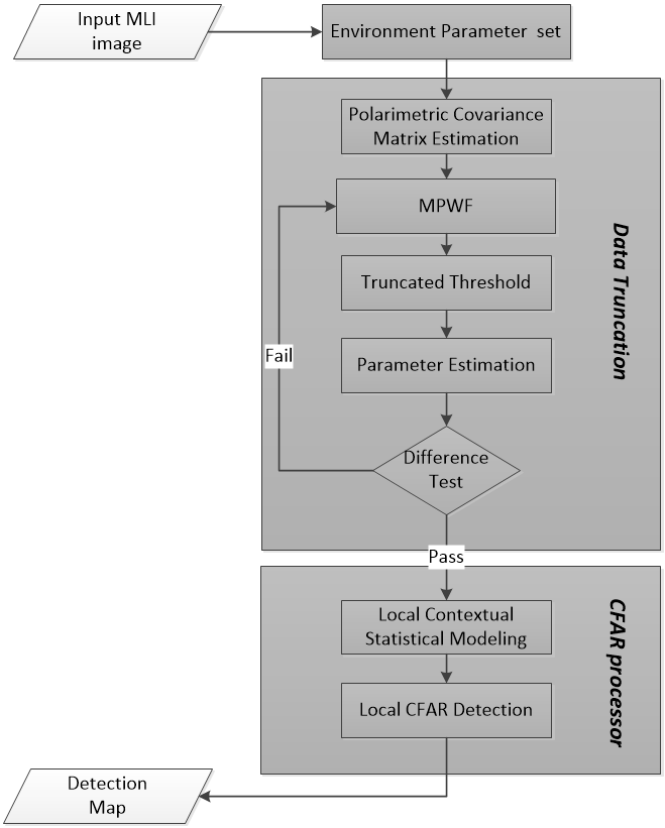


Fig. 5. Workflow of the Pol-TS-CFAR detector.

described in this section. The more challenging issues are the IC scheme and the adaptive setting of the truncation threshold. The workflow is indicated by solid arrow lines. The solid parallelograms represent the main input and output. The rectangular boxes within gray background regions show the operation and procedure, and the diamond is the decision of the termination of the iterative estimation.

A. Local Detection Design

To accelerate the speed of image processing, local block detection is proposed, which consists of dividing the whole image into blocks [35], [52].

As an initial step in our processing chain, we choose the local block threshold algorithm in the region of interest (ROI), where the ROI is small enough to be an adequate approximation for complicated large background as well. Based on the block design and suitable estimated parameters, the local CFAR threshold for each segmented ROI can be obtained using (27) or (40). In our experiments, only one image block is presented for the sake of brevity.

B. Iterative Censoring (IC) Scheme

The IC scheme was proposed by Barboy *et al.* [53] to solve the parameter estimation problem in the multiple-target situation. The threshold is iteratively updated based on the censored and remained reference samples. Samples get excluded from the censored reference samples when the specific samples

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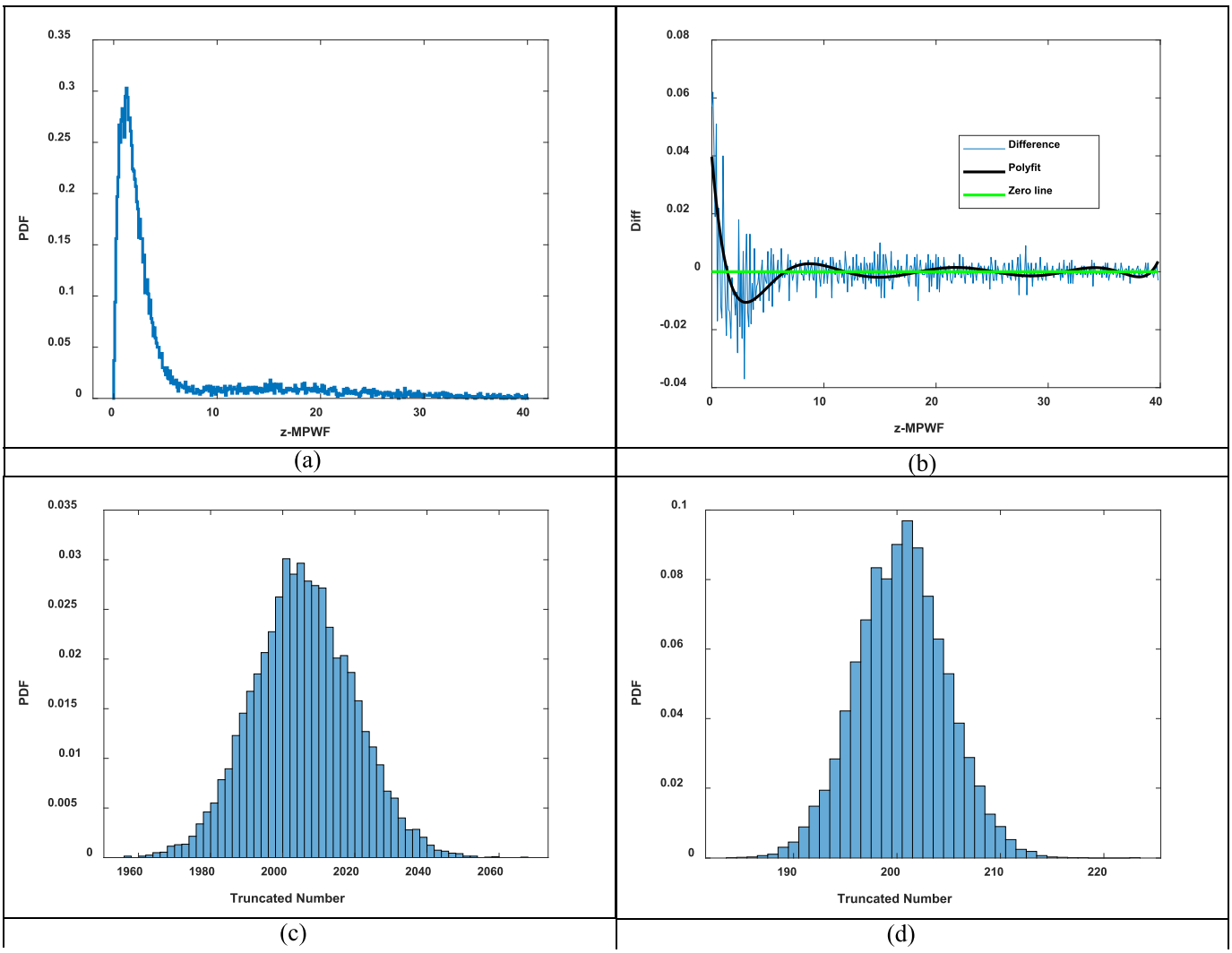


Fig. 6. Adaptive Determinant of truncated threshold. (a) PDF of the z-MPWF. (b) First-order difference of the PDF. (c) Truncated result (sample size 10 000). (d) Truncated result (sample size 1000).

exceed the adaptive threshold. This is repeated until there are no changes in both the threshold and the number of the reference samples [54], [55]. It is no doubt that multistep adaptive detection procedures may require many cycles and cost long calculation time. Despite this drawback, the IC scheme has shown robust performance in the multitarget situation and has been integrated within CFAR algorithms.

In our truncation processor, we first use all the pixels in the ROI to estimate the parameters and the polarimetric covariance matrix, followed by the MPWF. Then, we use the fixed or the adaptive setting of truncation threshold in Section IV-C. Then, we estimate the polarimetric covariance matrix and the other parameters by the remained pixels after truncation. Then, we use the new parameters to repeat the MPWF and the following steps until the truncated number is stable.

C. Determination of Truncated Threshold

The most important thing for the truncated processor is to separate the targets from sea clutter accurately after the truncation of the MPWF. We need to set the truncated threshold suitably. Note the fact that the PDF may be bimodal. One methodology to set the truncation is by smoothing the PDF and analyzing the saddle point. In this version, we do this manually. However, we will investigate the automatic methodologies exploiting multiemporal data for the future.

Here, we evaluate the difference of PDF and fit a smoother curve. We then look for the first minimum point. Simulations are performed to validate this methodology, as shown in Fig. 6. In the simulations, we first generate the sample sizes of the sea clutter (10 000) and the target (2000) with the designed polarimetric covariance matrices. Then, we plot the histogram of all these data after the MPWF as shown in Fig. 6(a). The first-order difference of the PDF is derived, and the poly function is used to fit a curve. Generally, the polynomial order is difficult to determine, which is set to 12 manually in our simulation, and therefore, the truncation threshold is 6.5. The truncated result is shown in Fig. 6(c). If we change the sample sizes of sea clutter and targets to be 1000 and 200, respectively, the performance is still excellent.

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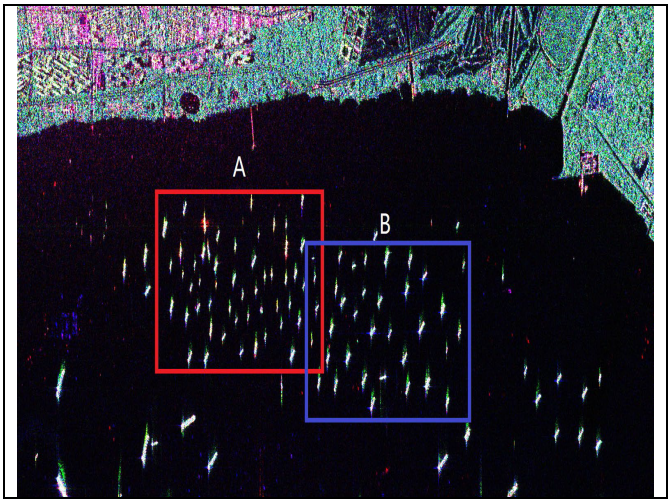


Fig. 7. C-band HH polarized SLC RADARSAT-2 SAR image acquired over the Singapore Strait area.

TABLE I
COMPARISONS OF THE ESTIMATIONS OF THE ENL (A)

	ENL_ML	ENL_SN	ENL_SM	ENL_GΓD	ENL_UT
L	2.2390	3.2732	3.3781	3.3086	2.1561
miu	1.0201	1.0067	1.0060	1.0182	1.0000

D. Test With Real Data

In this section, we test the goodness-of-fit of the statistical model for real data. Key methodologies presented in Sections IV-A–IV-C are applied here. Here, two data sets from different satellites are used here. We first use a C-band RADARSAT-2 polarimetric SLC SAR image acquired over the Singapore Strait area [56]. The image size is 1227 (slant range) × 2070 (azimuth) pixels. The nominal spatial resolution is 5.2 m (slant range) × 7.6 m (azimuth), and the pixel spacing is 4.7 m (slant range) × 4.8 m (azimuth). The incidence angle is about 47°. To perform validation, we use polarization information to assist in visually identifying small ships and remove azimuth ambiguities [57]. The final ground reference of ships marked with boxes is shown in [56], including 184. From counting the number of visible pixels of the ships, it appears that the vessels should approximately have lengths between 25 and 360 m. We choose ROI A and B randomly in Fig. 7 to test our methodologies. Therefore, there is no special reason for A and B to be partially stacked. We use the multilook to reduce speckles, and the nominal number of looks is 4. (The multilook processing is to suppress the speckles, while it will reduce the resolution of the SAR image. Generally, it is taken as 4 or 9. Here, we take it as 4.) Since the single look detector is a special to the multilook case, we only discuss the multilook case in our experiments.

After the multilook, the size of ROI A is 150 ×200. The IC scheme is performed and the results of the truncation are shown in Fig. 8. The truncated threshold is 6.5, which is derived by the minimum in Fig. 8(b).

The results of the parameter estimation are listed in Table I. The PDF of the truncated clutter is shown in Fig. 9.

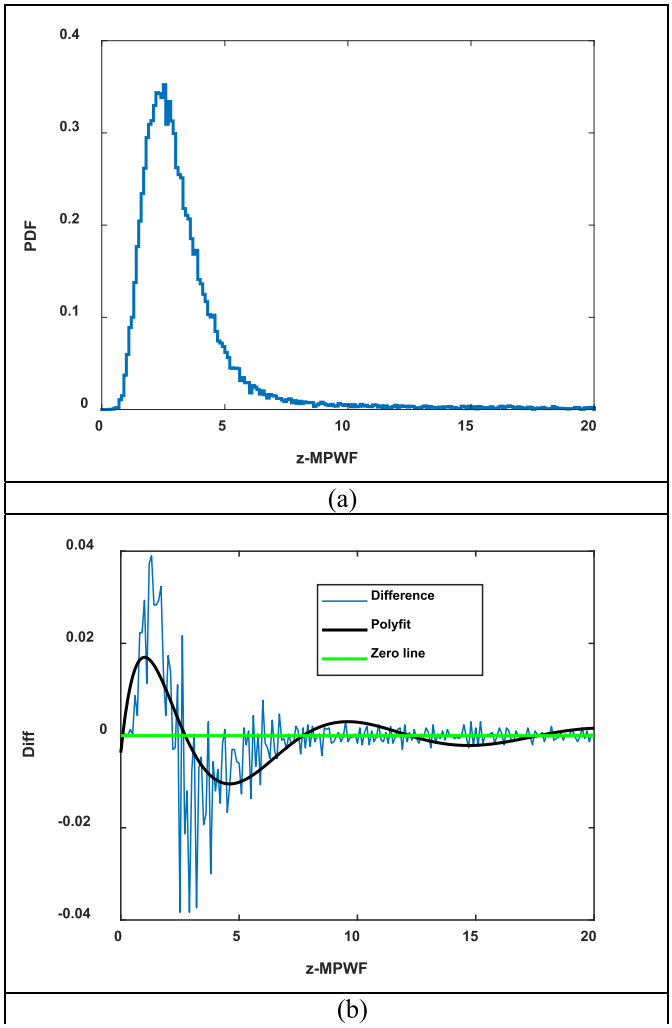


Fig. 8. Calculation of the truncation threshold in ROI A. (a) PDF in ROI A. (b) First-order difference of the PDF in ROI A.

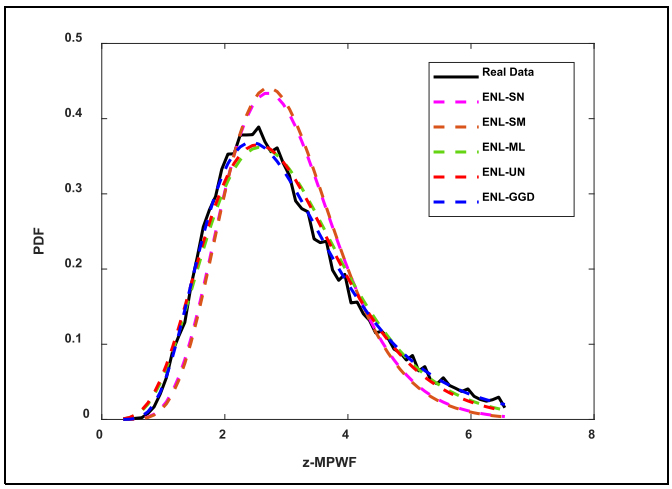


Fig. 9. PDFs of the truncated data after MPWF.

ENL-ML is estimated using the ML criterion, and the results derived from (20), (24), and (25) are almost the same after the IC scheme. Therefore, we use ENL-ML to describe all

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TABLE II
KL DISTANCE OF DIFFERENT PARAMETERS (A)

	ENL_ML	ENL_SN	ENL_SM	ENL_GΓD	ENL_UT
KL	0.0065	0.0413	0.0478	0.0016	0.0081

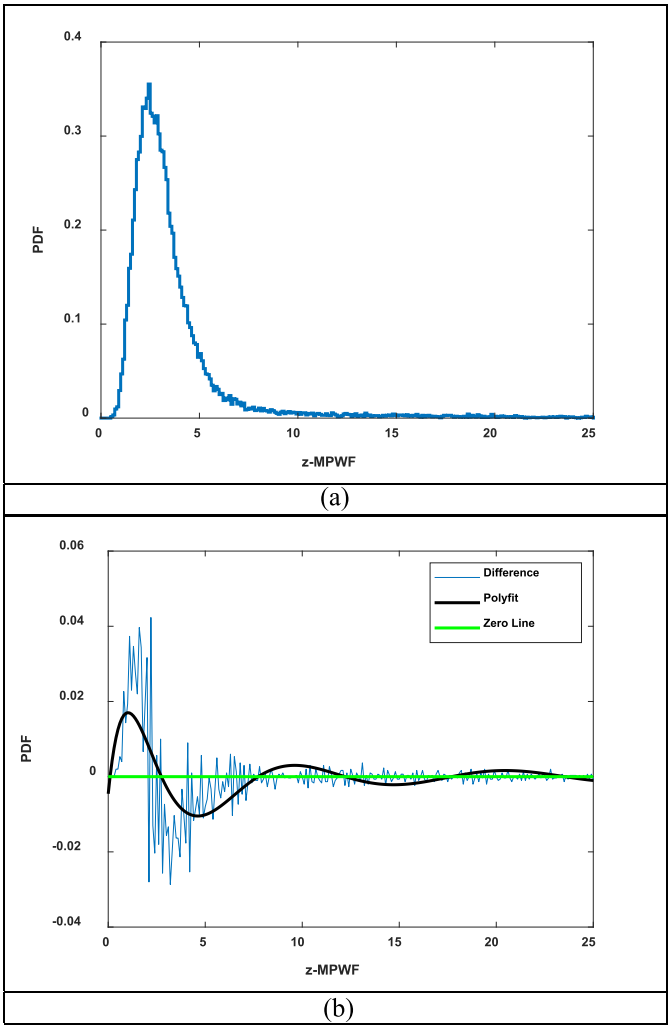


Fig. 10. Calculation of the truncation threshold in ROI B. (a) PDF in ROI B. (b) First-order difference of the PDF in ROI B.

the three ML estimators. ENL-SN is the estimator proposed by Anfinsen for the untruncated data, while processing the truncated data. Since the estimators based on the submatrices obtain almost the same results, we use ENL-SM to describe the submatrices estimators. ENL–GDD is solved by (34)–(36). ENL-UT is the estimator proposed by Anfinsen processing the untruncated data (not doing the preprocessing of target truncation).

The PDFs of different parameters estimated in Table I are shown in Fig. 9. It can be seen that the ENL-GGD gives the best performance, followed by the ENL-ML method.

In Table II, the KL distance [58] is used to measure the goodness-of-fit. The results in Table II show the same conclusion as that from Fig. 9. We can also see the ENL-ML give a good fit, where ENL is 2.239. If we estimate the ENL by a pure sea clutter area, the ENL should be 3.2. This means

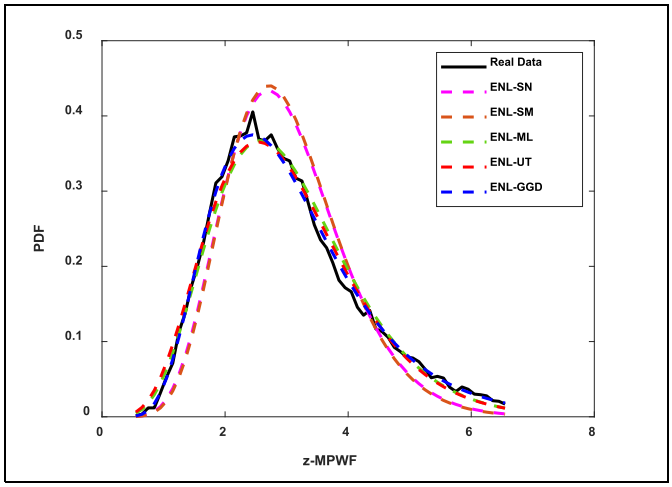


Fig. 11. PDFs of the truncated data after MPWF.

TABLE III
COMPARISONS OF THE ESTIMATIONS OF THE ENL (B)

	ENL_ML	ENL_SN	ENL_SM	ENL_GΓD	ENL_UT
L	2.3085	3.2630	3.3618	3.3080	2.1546
miu	1.0185	1.0068	1.0061	1.0160	1.0000

TABLE IV
KL DISTANCE OF DIFFERENT PARAMETERS (B)

	ENL_ML	ENL_SN	ENL_SM	ENL_GΓD	ENL_UT
KL	0.0059	0.0341	0.0397	0.0013	0.0078

TABLE V
COMPARISONS OF THE ESTIMATIONS OF THE ENL (C)

	ENL_ML	ENL_SN	ENL_SM	ENL_GΓD	ENL_UT
L	1.5595	2.7696	2.9024	2.7989	2.1248
miu	1.0301	1.0038	1.0032	1.0039	1.0000

the ENL estimated by the ENL-ML is to make the distribution of the clutter closest to the gamma model, while the ENL estimated here may not be the global one.

We repeat the experiments in ROI B. The results are shown in Figs. 10 and 11, and Tables III and IV. The size of ROI B is 200 $\times$ 150.

The conclusion in ROI B is the same as in ROI A. The results show that ENL-GGD estimator and ENL-ML estimator give better performance than the other estimators. In this data set, we can conclude that these two estimators give better detection performance than the others we tested.

The second data set is from the platform of GF-3, which also works in the C-band. The scene id is 3180124 and the image mode is QPSI, which means using "quad polarimetric strip 1" method to form image. The nominal number of looks is 4. The final ground truth is obtained by the same way addressed in the RadarSat-2 data set. Five objects are obtained in the ROI C. The biggest are oil platforms, and the others are ships. The ROI C is shown in Fig. 12.

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Fig. 12. C-band HH polarized SLC GF-3 SAR image acquired in South China sea.

TABLE VI
KL DISTANCE OF DIFFERENT PARAMETERS (C)

	ENL_ML	ENL_SN	ENL_SM	ENL_ GΓD	ENL_UT
KL	0.0080	0.0947	0.1117	0.0017	0.0304

The results of the statistics and parameter estimations are shown in Figs. 13 and 14 and Tables V and VI.

The truncation threshold is determined as 6.8 in ROI C.

The conclusion about different estimators is the same as that in ROI A and ROI B.

E. Detection Performance Analysis

To check the performance of the CFAR, we use the density-based clustering method named density-based spatial clustering of applications with noise (DBSCAN) [59]. We can validate that the single pixel detected in this image frame can be associated with false alarms. Please note this is not true in every data set where the size of ships can be smaller. Based on the previous visual inspection, we set the circle radial eps $= 25$ and minimum points MinPt $= 2$, which means two points in a circle with radius of 25 m can be seen as a ship. This applies a second layer of false alarm rejections. It should be noted that we use this clustering filter only to find points that can be associated with false alarms.

In addition, a figure of merit (FoM) is used to evaluate the detection performance [60]

$$\text{FoM} = \frac{N_{td}}{(N_{fa} + N_{gt})} \tag{45}$$

where N_{td} is the number of detected ships, N_{fa} is the number of false ships, N_{gt} is the number of real ships in the ROI. In the following figures, a red rectangle means an omitted target, a yellow rectangle means a false target, and a green rectangle means one true ship.

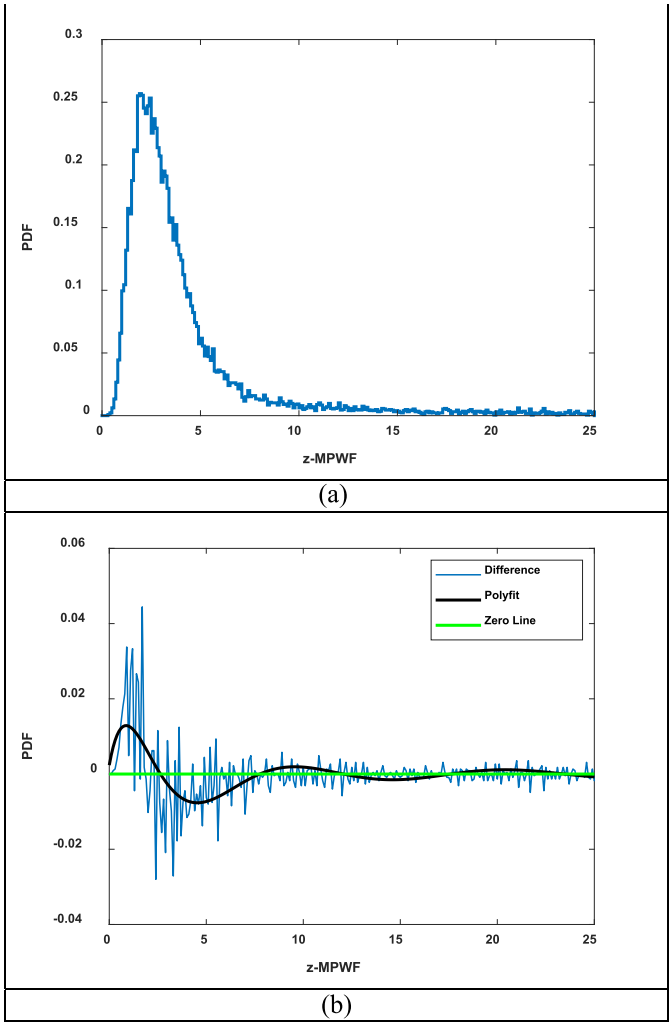


Fig. 13. Calculation of the truncation threshold in ROI C. (a) PDF in ROI C. (b) First-order difference of the PDF in ROI C.

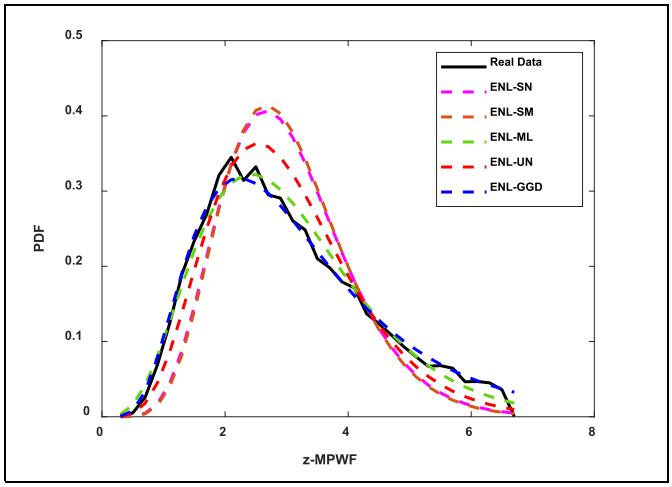


Fig. 14. PDFs of the truncated data after MPWF (ROI C).

We use the parameters estimated in Section IV-D to solve the threshold of the CFAR detection via (27) and (40), and the detection performances are presented as follows. The MPWF-$\times\times$ means the different estimators after MPWF,

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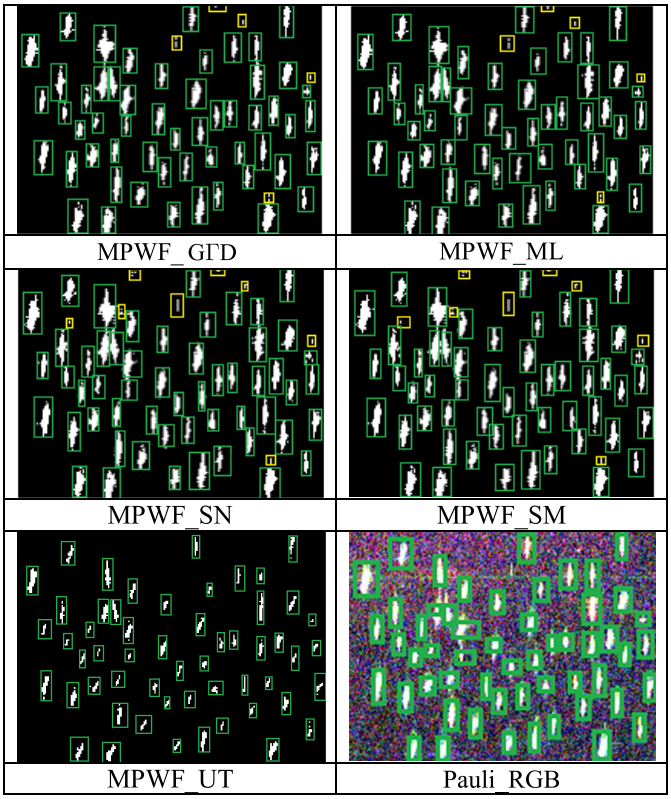


Fig. 15. Detection results of the CFAR detector based on MPWF (ROI *A*).

TABLE VII
PERFORMANCES OF ALL DETECTORS IN ROI *A*

Area	Method	$\hat{P}_{fa}$	N_{td}	N_{fa}	$FoM(\%)$
A	ENL_UT	0.0001	55	0	100.00
	ENL_ML	0.0005	55	5	91.67
	ENL_SN	0.0014	55	8	87.30
	ENL_SM	0.0014	55	8	87.30
	ENL_GΓD	0.0002	55	5	91.67

according to the ENL-×× methods. The MPWF-UT means the output of MPWF without data truncation. The ground truth is presented in the Pauli-RGB image.

The PFA is set as 0.0001. The results are presented as follows. The PFA and the FoM in ROI *A* are listed in Table VII.

The experiments are repeated in ROI *B*. The results are shown in Fig. 15 as follows.

The PFA and the FoM in ROI *B* are listed in Table VIII.

The results show that the two parameters ML (ENL-ML) have the same performance compared to the three parameters GGD (ENL-GGD). However, we can expect that if the sea clutter turns to be complicated the ENL-GGD may provide better performance than the ENL-ML. Although the ENL-UT gives the best FoM, it omitted one target, and from all the figures we can see it omitted many pixels of each ship.

Moreover, in Figs. 12 and 13 some of the detected false alarms are very small compared to the real targets, and they are close to the real targets. In other words, these false alarms are phenomena generated by ship sidelobes, or ship wakes; additionally other false alarms may be generated by azimuth ambiguities.

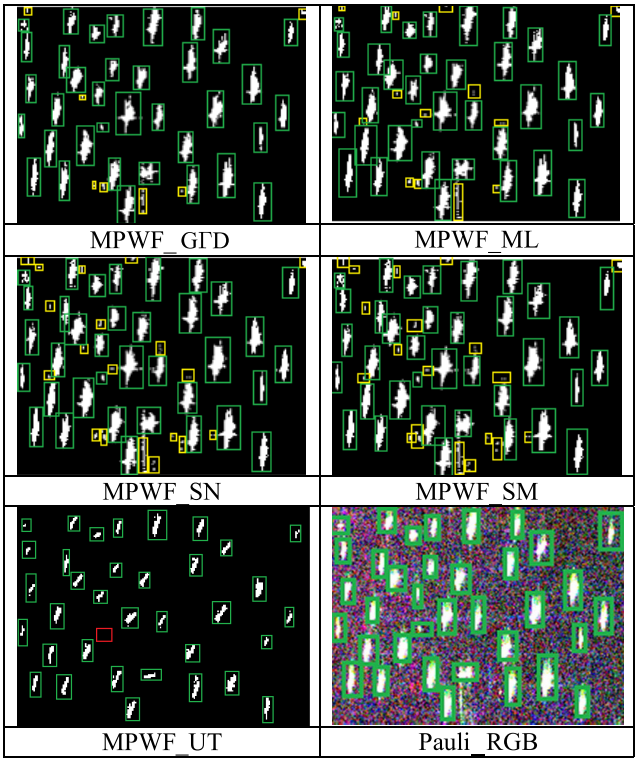


Fig. 16. Detection results of the CFAR detector based on MPWF (ROI *B*).

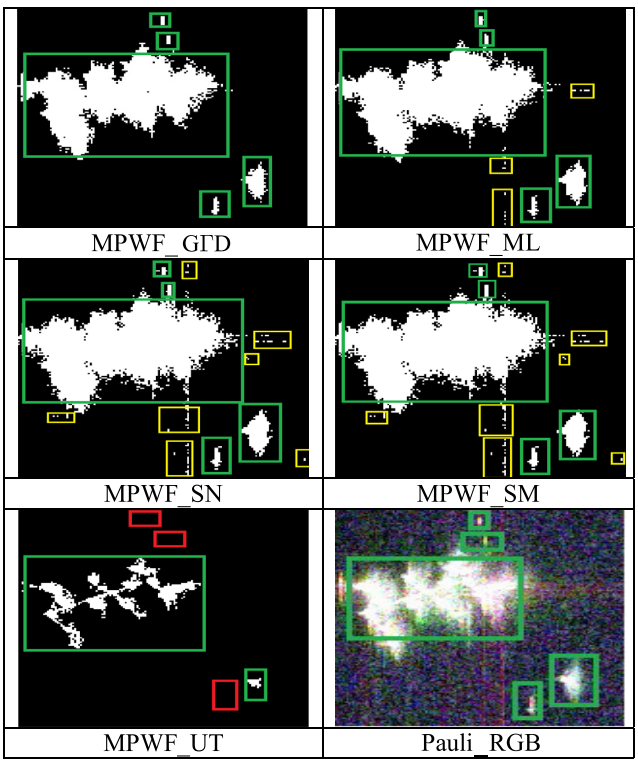


Fig. 17. Detection results of the CFAR detector based on MPWF (ROI *C*).

To validate the robustness to different satellites, the GF3 data are also tested. The experiments are repeated in ROI *C*. The results are shown in Fig. 17 and Table IX as follows.

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TABLE VIII
PERFORMANCES OF ALL DETECTORS IN ROI B

Area	Method	$\hat{P}_{fa}$	N_{td}	N_{fa}	$FoM(\%)$
B	ENL_UT	0.0002	35	0	97.22
	ENL_ML	0.0010	36	13	73.47
	ENL_SN	0.0017	36	19	65.45
	ENL_SM	0.0017	36	19	65.45
	ENL_GΓD	0.0002	36	7	83.72

TABLE IX
PERFORMANCES OF ALL DETECTORS IN ROI C

Area	Method	$\hat{P}_{fa}$	N_{td}	N_{fa}	$FoM(\%)$
C	ENL_UT	0.0001	2	0	40.00
	ENL_ML	0.0004	5	3	62.50
	ENL_SN	0.0014	5	7	41.67
	ENL_SM	0.0012	5	7	41.67
	ENL_GΓD	0.0002	5	0	100.00

The PFA and the FoM in ROI *C* are listed in Table IX.

Here, we can see the MPWF-GGD gives the best performance, but the MPWF-UT with no truncation omitted all the small ships, which has a weaker ability to detect small targets.

V. CONCLUSION

In this article, the PWF-TS-CFAR detector is proposed to improve the accuracy of estimated parameters for the clutter statistics in multiple target situations. The PWF is a very good method for ship detection when prior information of targets is absent. The PWF is applied to perform the truncation of PolSAR data. The estimation of the covariance matrix after truncation is derived in the Appendix. PWF-TS-CFAR does not require a guard window since the targets within the reference region is truncated before the clutter estimation. Therefore, it can also be used by a segmentation method [35] where there are no guard cells.

Since the degree of polarization, which has been reviewed in the introduction, provides very important information that is not covered by conventional covariance matrix optimization tools, such as the PWF detector, it would be good to compare the results we have obtained in this study with a new investigation that used the minimum degree of polarization in future.

APPENDIX

In this Appendix, we will derive the relation between the covariance matrix of the truncated data and that of the untruncated data following Tallis' method in [34]. From Tallis' study, it turns out that [34]

$$\mathfrak{S}(\rho;\boldsymbol{\Sigma}) = \frac{\boldsymbol{\Sigma}}{\mu_T(\rho)}, \quad \mu_T(\rho) \equiv \frac{F_d(\rho)}{F_{d+2}(\rho)} \tag{A.1}$$

with $F_v(\cdot)$ denoting the CDF of a χ^2 variable with d degrees of freedom. It is trivial to invert (18) so as to express $\boldsymbol{\Sigma}$ as a function of $\mathfrak{S}$. This is because $c_T(\rho)$ is a scalar damping factor independent of $\boldsymbol{\Sigma}$.This result is only suitable for real Gaussian-distributed vectors or covariance matrices.

The zero mean d-variate complex Gaussian under elliptical truncation is

$$p(\mathbf{x}) = \frac{1}{\pi^d|\boldsymbol{\Sigma}|}\exp(-\mathbf{x}^H\boldsymbol{\Sigma}^{-1}\mathbf{x}). \tag{A.2}$$

According to the methods in [34] and [47], and the relationship between the χ^2 cumulative distribution $F_v(\cdot)$ and the incomplete gamma function $\Gamma(a,b)$ [38], we can extended the real number results to the complex case

$$\boldsymbol{\Sigma} = \mu_T(\rho)\mathfrak{S}(\rho;\boldsymbol{\Sigma}), \quad \mu_T(\rho) \equiv \frac{\Gamma(d,\rho)}{\Gamma(d+1,\rho)} \tag{A.3}$$

with $\Gamma(a,b) = 1/\Gamma(a)\int_0^b y^{a-1}e^{-y}dy$.

The joint PDF of L independent samples based on (A.2) should be

$$p(\mathbf{x}_1,\mathbf{x}_2,\ldots,\mathbf{x}_L) = \frac{1}{\pi^{Ld}|\boldsymbol{\Sigma}|^L}\exp\left(-\sum_{i=1}^{L}\mathbf{x}_i^H\boldsymbol{\Sigma}^{-1}\mathbf{x}_i\right) \tag{A.4}$$

where $\boldsymbol{\Sigma}$ is positive semidefinite Hermitian, H is the conjunction transpose operator and defines a set $\mathfrak{D}$ as the limitation space as follows:

$$\mathfrak{D} = \{\mathbf{x}|0 \le \sum_{i=1}^{L}\mathbf{x}_i^H\boldsymbol{\Sigma}^{-1}\mathbf{x}_i = L\mathrm{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{C}) \le L\rho\} \tag{A.5}$$

where tr() is the trace operator, ρ is the truncation radius, and the five multilook polarimetric covariance matrix is $\mathbf{C} = 1/L\sum_{i=1}^{L}\mathbf{x}_i\mathbf{x}_i^H$.

Equation (A.4) can be seen as a general form of (A.2), and the only differences are the dimension of the vector and the polarimetric covariance matrix. We can denote $\mathbf{y} = (\mathbf{x}_1,\mathbf{x}_2,\ldots,\mathbf{x}_L)$, and then (A.2) turns to be

$$p(\mathbf{y}) = \frac{1}{\pi^{Ld}|\boldsymbol{\Sigma}|^L}\exp\left(-\mathbf{x}_i^H\boldsymbol{\Xi}^{-1}\mathbf{x}_i\right) \tag{A.6}$$

where

$$\boldsymbol{\Xi} = \begin{bmatrix} \boldsymbol{\Sigma} & 0 & \ldots & 0 \\ 0 & \boldsymbol{\Sigma} & \ddots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \ldots & \boldsymbol{\Sigma} \end{bmatrix}$$

is the Ld dimension covariance matrix.

Obviously, (A.6) is the same form as (A.2). The difference between them is the dimension. Therefore, we get the second moments of the truncated matrix $\Theta(\rho;\boldsymbol{\Sigma})$ as

$$\boldsymbol{\Xi} = \mu_T(\rho)\Theta(\rho;\boldsymbol{\Sigma}), \quad \mu_T(\rho) \equiv \frac{\Gamma(Ld,\rho)}{\Gamma(Ld+1,\rho)}. \tag{A.7}$$

Observing the form of the Ld dimension polarimetric covariance matrix is a block diagonal matrix, we can get the second moments of the truncated matrix $\mathfrak{S}(\rho;\boldsymbol{\Sigma})$ as

$$\boldsymbol{\Sigma} = \mu_T(\rho)\mathfrak{S}(\rho;\boldsymbol{\Sigma}), \quad \mu_T(\rho) \equiv \frac{\Gamma(Ld,\rho)}{\Gamma(Ld+1,\rho)}. \tag{A.8}$$

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